

GEOG 172

Centrography

A spatial analogy to the statistical idea of the mean is the geographic center.

Lecture 2-1

IMPORTANT IDEAS

- Standard statistics sometimes have spatial analogs that are useful as simple summaries of spatial patterns.
- The spatial center is an important idea, but it can be calculated several different ways

TUTORIAL PREVIEW

We will ask you to use the descriptive spatial statistics learned in class to consider where the divide between NorCal and SoCal is. We will provide you with a starter notebook then ask you to fill in sections of code and complete short answer questions.

ACTIVITY

Overview: The mean center and standard distance are descriptive spatial statistics analogous to the mean and standard deviation. You will work to summarize the class GeoGuesser results using these statistics.

Purpose: Familiarize yourself with different spatial centers

Task:

- Calculate the mean center
- Calculate the weighted mean center

ACTIVITY

Task: Calculate the mean center using the following formula

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} \qquad \bar{y} = \frac{\sum_{i=1}^n y_i}{n}$$

Steps:

- Calculate the mean of all x-coordinates
- Calculate the mean of all y-coordinates
- Combine the two results to form a coordinate set for the mean center
- Plot that location on your map

ACTIVITY

Task: Calculate the weighted mean center using the following formula

$$\overline{x_w} = \frac{\sum_{j=1}^n w_j x_j}{\sum_{j=1}^k w_j} \quad \overline{y_w} = \frac{\sum_{j=1}^n w_j y_j}{\sum_{j=1}^k w_j}$$

Steps:

- Multiple each guess coordinate by the population weight
- Apply the remainder of the formula
- Plot that location on your map

ACTIVITY

Discussion:

1. Are the two centers in the same location? If they are different, why are they?
2. Which calculation preserves more information from the original guesses?
3. What would happen if we used larger regions? What if we used smaller regions?

STANDARD DISTANCE

You can also calculate the standard distance and weighted standard distance of these points and regions using the following equations.

$$SD = \sqrt{\frac{\sum_{i=1}^n w_i d_{ic}^2}{\sum_{i=1}^n w_i}}$$

For the sake of time, I will leave this to you to try

GEOGRAPHIC CENTER

You can calculate geographic centers for regions (polygons) as well as points.

$$x_c = \frac{1}{6A} \sum_{i=0}^{n-1} (x_i + x_{i+1}) (x_i y_{i+1} - x_{i+1} y_i)$$

$$y_c = \frac{1}{6A} \sum_{i=0}^{n-1} (y_i + y_{i+1}) (x_i y_{i+1} - x_{i+1} y_i)$$

with area, A , of the polygon found using

$$A = \frac{1}{2} \sum_{i=0}^{n-1} (x_i y_{i+1} - x_{i+1} y_i)$$

SPHERICAL CENTER

If you assume the earth is spherical (it almost is) you can calculate the **weighted spherical center** using

Step 1:

Convert each latitude-longitude to a 3D Cartesian coordinate

$$x = \cos \lambda \cos \varphi, \quad y = \sin \lambda \cos \varphi, \quad z = \sin \varphi$$

where φ is latitude and λ is longitude.

SPHERICAL CENTER

If you assume the earth is spherical (it almost is) you can calculate the **weighted spherical center** using

Step 2:

Calculate the weighted mean center of each coordinate

$$\overline{x}_w = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^k w_i} \quad \overline{y}_w = \frac{\sum_{i=1}^n w_i y_i}{\sum_{i=1}^k w_i} \quad \overline{z} = \frac{\sum_{i=1}^n w_i z_i}{\sum_{i=1}^k z_i}$$

SPHERICAL CENTER

If you assume the earth is spherical (it almost is) you can calculate the **weighted spherical center** using

Step 2:

Transform back into latitude and longitude

$$\sin^{-1}\left(\frac{\bar{z}}{t}\right), \quad \tan^{-1}\left(\frac{\bar{y}/t}{\bar{x}/t}\right), \quad (1 - t) R$$

where

$$t = \sqrt{\bar{x}^2 + \bar{y}^2 + \bar{z}^2}$$

and R is the radius of the earth, $R = 3,953$ miles.

The Center of the U.S.

From concept to computer

DEMONSTRATION

1. Where is the center of the U.S.?
2. How has the center of the U.S. moved over time?
3. Does the method used seem to alter the location of the center?
4. How dispersed is the U.S. population?